

DAY -1 : INTRODUCTION TO HARDWARE THEORY

1. Diagram of Wifi Setup



2. Main Functions of Networking Hardware Devices

1. Modem (Modulator-Demodulator)

- **Main Function:** Converts incoming signals from your Internet Service Provider (ISP)—such as fiber, cable, or DSL—into digital data that your devices can read, and converts outgoing digital data back into analog signals.
- **Real-Life Example:** At home, connected near the wall outlet where the cable/fiber enters, providing raw internet into your house.

2. Router

- **Main Function:** Connects your local network to the internet, manages network traffic, assigns local IP addresses (via DHCP), and acts as a security firewall.
- **Real-Life Example:** At home, sitting between the modem and your devices to share one internet connection across multiple phones, TVs, and laptops.

3. Switch

- **Main Function:** Connects multiple wired devices together on the same local network, expanding the number of available physical Ethernet ports.
- **Real-Life Example:** In an office or gaming desk setup, used to plug a PC, console, smart TV, and network printer into the network using cables.

4. **Access Point (AP)**

- **Main Function:** Broadcasts a wireless signal (Wi-Fi) from a wired network line, extending wireless coverage to areas that are out of reach of the main router.
- **Real-Life Example:** Mounted on a hallway ceiling or second floor of a large house to ensure strong Wi-Fi coverage everywhere.

5. **Network Cable**

- **Main Function:** Serves as the physical medium to carry high-speed digital data signals between devices with low latency and high stability.
- **Real-Life Example:** Plugged directly between a desktop computer and a router for fast, lag-free online gaming or video conferencing.

3. **Types of Network Cables & Sources**

1. **Ethernet Cable (Twisted Pair - Cat5e / Cat6 / Cat6a)**

- **Common Use:** Used inside homes and offices to connect computers, routers, switches, and gaming consoles together in a Local Area Network (LAN).
- **Source:** [Wikimedia Commons - Ethernet Cable](#)

2. **Coaxial Cable (Coax)**

- **Common Use:** Used by broadband cable companies to deliver television signals and high-

speed internet from the street directly to your cable modem.

- **Source:** [Wikimedia Commons - Coaxial Cable Cutaway](#)

3. **Fiber Optic Cable**

- **Common Use:** Used for ultra-fast long-distance data transmission (using light pulses) for ISP backbone networks and Fiber-to-the-Home (FTTH) internet lines.
- **Source:** [Wikimedia Commons - Fiber Optic Cable](#)

4. [Comparison Table: Router vs. Switch](#)

<u>Feature / Metric</u>	<u>Router</u>	<u>Switch</u>
<u>Main Use</u>	<u>Connects different networks together (e.g., your local home network to the outer internet).</u>	<u>Connects multiple devices on the same local network together (expands wired ports).</u>
<u>Typical Location</u>	<u>Situated at the gateway/edge of your network, directly connected to the modem.</u>	<u>Placed within the local network, connected downstream to one of the router's ports.</u>
<u>Data Handling</u>	<u>Uses IP Addresses (Layer 3) to route data packets between internal devices and the internet.</u>	<u>Uses MAC Addresses (Layer 2) to deliver data directly to a specific connected device.</u>
<u>Network Boundary</u>	<u>Assigns internal IP addresses (DHCP) and creates a protective firewall barrier.</u>	<u>Does not assign IP addresses or manage security; simply passes data locally between ports.</u>

